21/04/2026

Actividad 3.3

Trayecto 1

Exponencial

$x = [0 \text{ a } 8]$

$f(x) = e^{0.2t}$

Condiciones iniciales

$x(0) = 0$

$y(0) = e^{0.2(0)} = 1$

$\theta(0) = \tan^{-1}(0.2e^{0.2(0)}) = \tan^{-1}(0.2)$

$x(t) = t$

$y(t) = e^{0.2t}$

$\dot{x}(t) = 1$

$\dot{y}(t) = 0.2e^{0.2t}$

$\theta(t) = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right) = \tan^{-1}\left(\frac{0.2e^{0.2t}}{1}\right)$

Vel lineal

$v(t) = \sqrt{\dot{x}^2 + \dot{y}^2} = \sqrt{1^2 + (0.2e^{0.2t})^2}$

Vel angular

$\omega(t) = \frac{d\theta}{dt} = \frac{d(\tan^{-1}(\overset{dx\ 0.04e^{0.2t}}{0.2e^{0.2t}}))}{dt} = \frac{0.04e^{0.2t}}{1 + (0.2e^{0.2t})^2}$

Trayecto 2

Cosenoidal

$x = [8 \text{ a } 18]$

$f(x) = 3\cos(0.4(t-8))$

$x(t) = t$

$y(t) = 3\cos(0.4(t-8))$

$\theta(t) = \tan^{-1}(-1.2\sin(0.4(t-8))$

$\dot{x}(t) = 1$

$\dot{y}(t) = -1.2\sin(0.4(t-8))$

Vel lineal

$v(t) = \sqrt{1^2 + (-1.2\sin(0.4(t-8)))^2}$

Vel angular

$\omega(t) = \frac{d(\tan^{-1}(\overbrace{-1.2\sin(0.4(t-8))}^{dx}))}{dt}$

$\omega(t) = \frac{-0.48\cos(0.4(t-8))}{1 + (-1.2\sin(0.4(t-8))^2}$

Trayectoria 3

Logarítmica

$x = [18 \text{ a } 30]$

$f(x) = 2\ln(t-17)$

$x(t) = t$

$y(t) = 2\ln(t-17)$

$\theta(t) = \tan^{-1}\left(\frac{2}{t-17}\right)$

$\dot{x}(t) = 1$

$\dot{y}(t) = \frac{2}{t-17}$

Vel lineal

$$U(t) = \sqrt{1^2 + \left(\frac{2}{t-17}\right)^2}$$

Vel angular

$$\omega(t) = \frac{d\left(\tan^{-1}\left(\frac{2}{t-17}\right)\right)}{dt} = \frac{\frac{-2}{(t-17)^2}}{1 + \left(\frac{2}{t-17}\right)^2}$$